

Solutions to Assignment 3

Advanced Tools for Complex Network Analysis

CNET 5052, Spring 2026

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1. *Question 1.*

(a) See below.

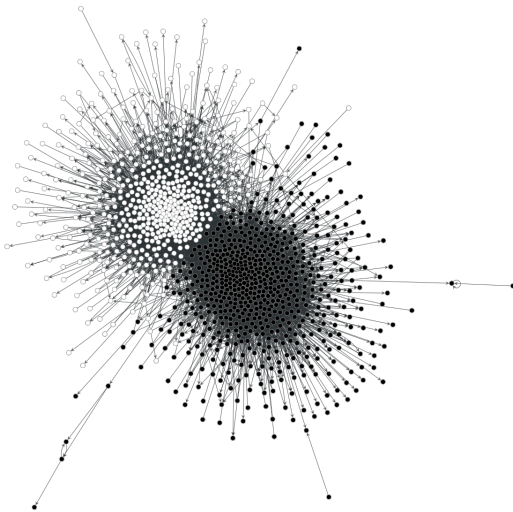


FIG. 1. Degree-Corrected

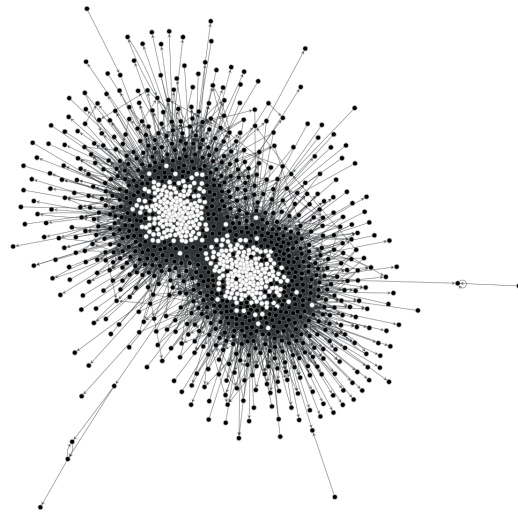
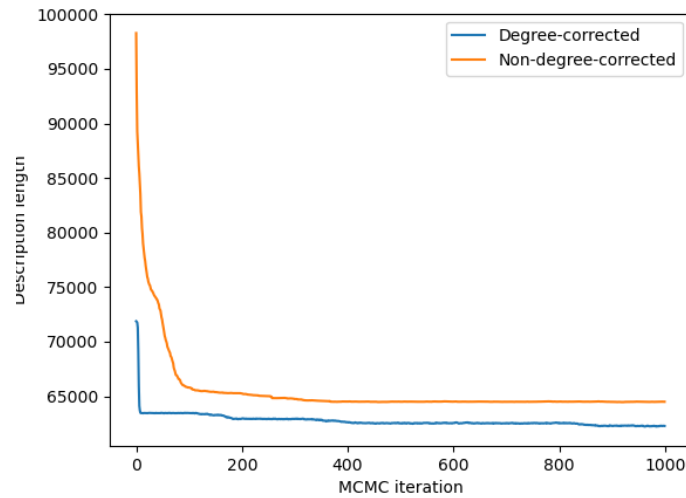


FIG. 2. Non-Degree-Corrected

- (b)
- i. Degree-corrected: 63427.54 Non-degree-corrected: 83131.51
 - ii. The degree-corrected model converges to a much lower description length, as well as doing so faster. This is because the non-degree-corrected model can't explain degree variation within blocks, so it's paying a penalty for this unexplained structure.

* ADD CLASS REPO FORK <https://github.com/caleb-chandler/cnet5052sp26>



- (c) Degree correction is a solution to the problem of the baseline SBM assuming nodes in the same block are statistically equivalent (same expected degree). It does this by adding a term θ_i for node i 's propensity to form edges relative to others in its block. This value is then used to form the equation $E[A_{ij}] = \theta_i \theta_j \omega_{z_i z_j}$, which recovers the expected number of edges between two nodes.

2. Question 2.

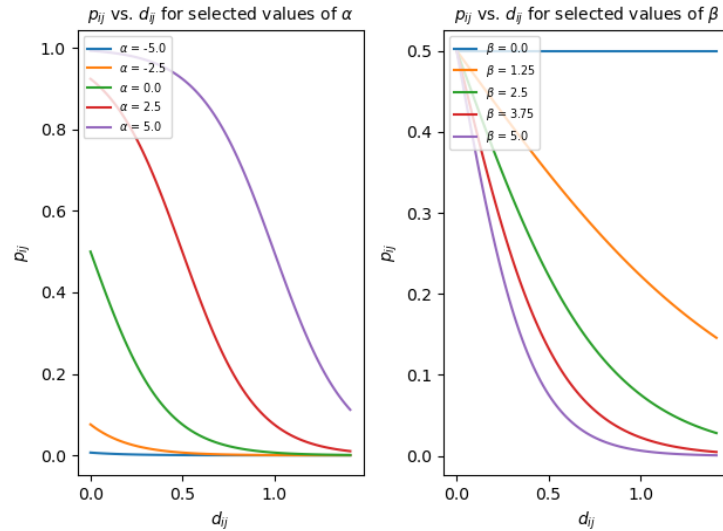
- (a) My network features are degree, clustering, closeness, and PageRank. For the synthetic features I did tenure and height. For Mr. Hi, I did mean 360 and std dev 55 for tenure (as in days) and 175/7 in cm for height. Officer was 290/73 for tenure and 181/8 for height. I chose these values to sound plausible for what I'm hypothetically measuring.
- (b) Shown below.
- (c) Shown below. I used the default sklearn metric. The model is primarily using my synthetic features, which probably says less about the predictive power of these features than it does about the weakness of the network ones. One of the first things that jumps out to me in thinking about this is the challenge of community detection, which I have gained a new appreciation for in light of this result.
- (d) I chose to increase noise. I had to 3x it to start getting a notable difference, which is due to the choices of distribution parameters. Tenure already has a lot of noise to begin with, so an increase in noise doesn't penalize it as significantly, and Height has such *little* noise that it's not as sensitive to multiplicative increase. In any case, 3x-ing noise causes both Accuracy and F1 to drop off significantly. Predictably, this performance reduction is almost entirely driven by Height, as the model was relying on the low std dev for nearly all of that feature's predictive power.

Baseline			Increased Noise (3x)		
Mean Accuracy 0.71			Mean Accuracy 0.61		
Mean F1 0.70			Mean F1 0.60		
Std Accuracy 0.16			Std Accuracy 0.16		
Std F1 0.17			Std F1 0.17		
Pred. Mr. Hi Pred. Officer			Pred. Mr. Hi Pred. Officer		
Mr. Hi	2	1	Mr. Hi	2	1
Officer	2	2	Officer	1	3
Feature	Importance		Feature	Importance	
Tenure	0.340		Tenure	0.333	
Height	0.281		PageRank	0.178	
PageRank	0.137		Height	0.171	
Closeness	0.101		Clustering	0.123	
Clustering	0.089		Closeness	0.119	

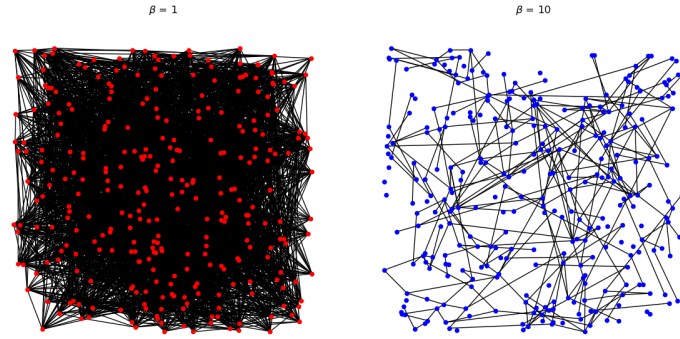
TABLE I. Random Forest classification results: baseline vs. increased synthetic feature noise.

3. Question 3.

- (a) Shown below. $\beta = 0$ causes the probability to remain uniform regardless of distance. This is because it cancels out distance altogether, so there is no variable parameter for p_{ij} to change with.



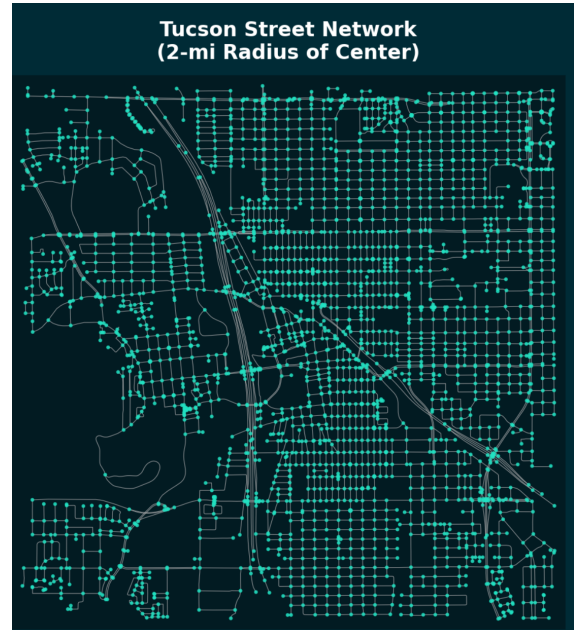
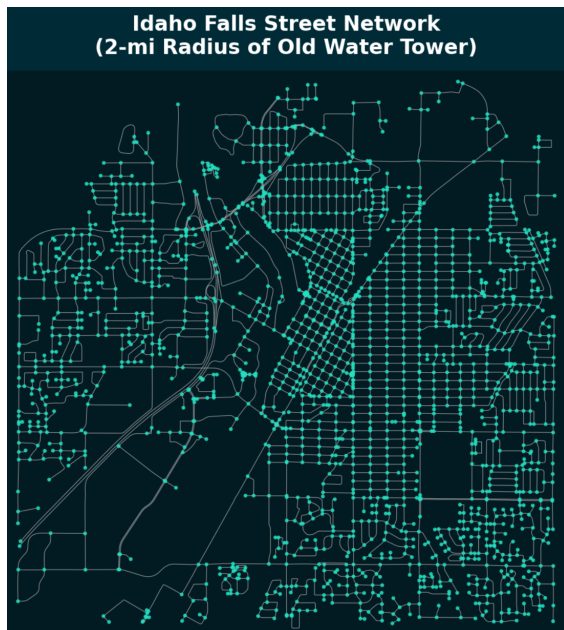
- (b) See notebook.
- (c) Small β : # Edges: 3397; Avg Degree: 22.647; Avg Distance (Edge Length): 0.454
 Large β : # Edges: 281; Avg Degree: 1.873; Avg Distance (Edge Length): 0.181



- (d) It reduces the average edge length, which makes perfect sense; the average edge length will naturally drop off with a greater bias against the formation of long distance connections.

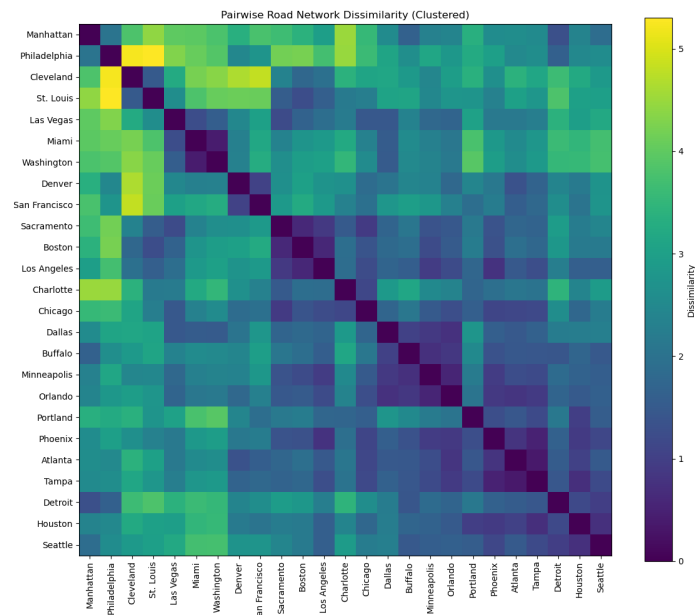
4. Question 4.

- (a) I chose the undirected simple graph representation. For the city I picked my hometown (Idaho Falls), and centered it on a long-standing landmark that was controversially torn down recently. # Nodes: 1847 — # Edges: 2688
- (b) OSMnx couldn't read the coordinates, so I had to just put in Tucson and let it decide the center. # Nodes: 2781 — # Edges: 4384



- (c)
- Tucson is of course much bigger; the N and M results show that it's also significantly denser at the center.
 - Both cities largely maintain a grid layout, though Tucson much more so.
 - Tucson has much shorter edge lengths due to being more compact.

- iv. Almost the entirety of Idaho Falls is contained in this radius, whereas it only seems to capture a small portion of Tucson.
 - v. Idaho Falls has a major river running through it (hence the "falls"), so you can see a pretty clear division between sides.
- (d) The first thing that stands out to me as being a meaningful/fun differentiator for road networks is what you might call "gridness." This can be measured by several features; the ones I'm choosing are clustering coefficient, std dev of degree, and std dev of edge lengths.
- (e) All of these features measure gridness in different ways. Clustering coefficient in a grid structure should be low, since closed triangles would break the pattern. The other two are more straightforward: Degree should be relatively homogeneous in a grid, as should the lengths of roads between intersections. All three features are scale-invariant as to the network size (degree usually wouldn't be, but this is a physical network with local constraints on intersection size), but they are not themselves measured on the same scale. As a result, comparisons between just two graphs will be dominated by std dev of edge length because it is measured in meters while the other two features are close to zero in magnitude. Normalizing with only two graphs doesn't help: z-scoring maps both values to ± 1 for every feature, collapsing all differences to the same magnitude and erasing any information about which features actually differ more. The 25-city comparison, however, is a different story: z-score normalization uses meaningful population-level means and standard deviations, so all three features contribute proportionally to the distance measure.
- (f) Distance is 37.276. In light of the normalization challenge for two-graph comparisons, this result is dominated by edge length. Nevertheless, it still aligns with my observation of Tucson having shorter such lengths.
- (g) To make my visualization look nicer and be more interpretable, I used hierarchical clustering to group similar cities together.



If you reference anything, add it to the bibliography below and cite it in the text using `\cite{Hartle2020_wegd}` [1].

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- [1] Hartle, H., Klein, B., McCabe, S.D., Daniels, A., St-Onge, G., Murphy, C., & Hébert-Dufresne, L. (2020). Network comparison and the within-ensemble graph distance. *Proceedings of the Royal Society A*, 476: 20190744. doi: [10.1098/rspa.2019.0744](https://doi.org/10.1098/rspa.2019.0744).